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Want a high-flying career? The booming field of avionics beckons you

Samir Makwana

Like the heart is an integral part of the human body, an aircraft also has a heart—the avionics system. This system forms the integral part of an aircraft design and increases its capability. All the electronic components utilised in an aircraft for display, communication, navigation and management of other systems linked with them form the avionics system. Electronics systems like communications, surveillance and navigation systems, display systems, flight management systems, engine, fuel systems and other primary functions are used on an aircraft.

Avionics systems play a crucial role for effectiveness of the aircraft. Complex as well as expensive by nature, avionics systems are often called 'systems of systems'. For designing, integrating and testing the avionics systems and subsystems, a lot of expertise, time and money is required.

For the past five years, the Indian aerospace and aviation industry has been registering significant growth. Praful Patel, Minister of State for Civil Aviation, stated at the World Economic Forum held at Davos in January that the current annual growth of numbers of domestic passengers is 40 per cent and an average growth of 25 per cent per year is expected over the next five years.

Broader Scope

India being one of the fastest growing markets for civil aviation, leading avionics companies have a presence in this country directly or indirectly. For any outsourcing business like design, development or production, India has always served to be the ideal option. "Avionics jobs are similar to other engineering jobs, with the exception of need for aircraft knowledge", says C. V. Ravi Shankar, Director, Merlin Hawk Aerospace, involved in research, design and value added development of hardware and software for aviation and aerospace applications.

Civil and defence are two areas you can work in. The functional roles remain nearly the same in both these areas. An avionics candidate can be involved in functional roles like design, development, integration, testing and advanced research of the hardware and software for avionics systems.

The private sector in the avionics field comprises the aerospace components suppliers / OEMs, IT solutions providers and other small and medium enterprises dealing with maintenance and allied services. Speaking of IT specifically, top IT companies like Infosys, Satyam, HCL, TCS, Wipro, Hindustan Aeronautics Limited, to name a few, specialise in providing solutions as aerospace IT players.

"In avionics, one should know and be sure of where and what is the specialisation required



Chaitanya Surpur